

Paper Review: The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games

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Condensed conclusions, limitations, implementation lessons, and experiments to consider

Scope of this report: this paper informs the MARL training backbone and evaluation discipline. It does not, by itself, decide the full direction of the emergent-communication project.

1. What the Paper Is Mainly About

The paper asks whether PPO, an on-policy policy-gradient method often assumed to be sample-inefficient in MARL, can be a strong cooperative multi-agent baseline when implemented carefully. It does not propose a fundamentally new MARL algorithm. Instead, it studies PPO across multiple cooperative benchmarks, distinguishes centralized-critic PPO (MAPPO) from fully local PPO (IPPO), and isolates practical implementation choices that strongly affect performance.

- MAPPO: decentralized actors use local observations at execution, while the critic may use privileged/global information during training (CTDE).
- IPPO: both actor and critic use local information; no centralized value input.
- Main empirical claim: carefully configured PPO is often competitive with or superior to strong off-policy MARL methods in both final performance and sample efficiency.
- Benchmarks: Multi-Agent Particle Environment (MPE), StarCraft Multi-Agent Challenge (SMAC), Google Research Football (GRF), and Hanabi.
- The paper's most practically valuable contribution is its ablation study of five factors: value normalization, critic/value input representation, training-data reuse, PPO clipping, and batch size.

2. Core Results and Conclusions

Domain	Compared against	Main conclusion
MPE	QMIX, MADDPG	MAPPO is comparable/superior; IPPO is also surprisingly strong. MAPPO is slightly better than IPPO in some partially observable tasks.
SMAC	QMIX, RODE, QPLEX, CW-QMIX, AI-QMIX	MAPPO/IPPO solve most maps strongly; MAPPO is competitive with major off-policy baselines at comparable sample counts.
GRF	QMIX, CDS, TiKick	PPO clearly outperforms QMIX across tested scenarios and is competitive with stronger specialized methods.
Hanabi	SAD, VDN	MAPPO is competitive/superior, and the advantage of centralized critic information becomes clearer as player count grows.

Important interpretation: the paper does NOT show that centralized critics are always necessary. IPPO is already very strong in many tasks. It shows that centralized critic information can become particularly useful as partial observability / multi-agent complexity increases.

3. MAPPO / CTDE: What Matters

Actors and critic are separate neural networks. Each actor selects an action from the agent's local observation. The critic estimates the expected future return and exists only during training, so it may use information unavailable to the deployed actor.

- Actor input: local observation (and, in our future communication experiments, received messages/history if explicitly provided).
- Critic input: can contain global environment state plus useful agent-specific features.
- Execution remains decentralized: the critic is not required when the trained agents act.
- Why centralization can help: a better-informed critic can estimate values more accurately, reducing variance in the policy-learning signal.

4. The Five Major Implementation Findings

4.1 Value normalization

Value targets can vary greatly during MARL training. The authors standardize the critic's targets using running estimates of their mean and standard deviation, and denormalize critic outputs when needed for return/advantage calculations. This stabilizes value learning; in their ablations it never hurt and often significantly improved performance.

- Implementation takeaway: include running value-target normalization (the paper uses PopArt in recurrent MAPPO).
- Reason to test/monitor: reward scales may vary sharply across scenarios, curricula, or added communication penalties.

4.2 Critic/value input representation

Input	Meaning	Paper finding
IND	Only agent-local observation	Decentralized critic; may miss information needed for accurate value estimation.
CL	Concatenate all agents' local observations	Can become very high-dimensional and may still miss globally unobserved information.
EP	Environment-provided global state	Useful, but may omit agent-specific details.
AS	EP global state + current agent's local observation	Strong: combines global and local/agent-specific information.
FP	AS with redundant overlapping features pruned	Often strongest/cleanest; avoids needless dimensionality.

- Implementation takeaway: if the environment exposes a global state, construct a critic input containing global information plus agent-specific features; remove duplicated features where practical.
- Experiment to consider: local critic vs global critic vs global+agent-specific critic. This is an ablation/supporting experiment, not the communication question itself.

4.3 Training-data usage: epochs and mini-batches

PPO first collects a rollout (fresh interaction data generated by the current policy), computes return/advantage targets, and then reuses that rollout for a limited number of gradient-training epochs. The paper finds that excessive sample reuse is harmful in harder MARL tasks, likely because all agents' policies are changing and the rollout becomes stale relative to the current policies.

- Paper practice: about 15 epochs on easy tasks; 10 or 5 on difficult tasks.
- Mini-batches: the paper generally avoids splitting the rollout; at most two mini-batches. In selected SMAC ablations, 1 mini-batch was best on 22/23 maps, while 4 mini-batches failed on all selected maps.
- Important clarification: epoch count is NOT episode count or number of maps. The environment can keep generating randomized episodes; 'epochs' means how many times the same recently collected rollout is reused for gradient updates.
- Implementation takeaway: collect fresh rollout -> calculate targets -> train a few epochs -> discard rollout data -> collect fresh rollout. Learned network weights are retained.

4.4 PPO clipping

PPO limits how drastically action probabilities and value predictions are allowed to change during an update. The clipping parameter epsilon controls update aggressiveness: larger epsilon permits bigger changes. The authors hypothesize that conservative updates are especially important in MARL because rapid changes by many agents increase non-stationarity.

- Paper finding: epsilon = 0.05 often learns more slowly but is stable and reaches strong final performance; large values such as 0.2-0.5 often produce suboptimal performance on tested SMAC maps.
- Paper suggestion: keep epsilon below 0.2 and tune the stability-vs-speed tradeoff.
- Implementation takeaway: use conservative PPO updates and treat clip ratio as a tuned hyperparameter rather than blindly importing a single-agent default.

4.5 Batch size

Batch size is the amount of fresh on-policy trajectory data collected before an update. Larger batches can give more reliable gradient estimates, but unnecessarily large batches require more compute/parallelism and can reduce sample efficiency.

- Implementation takeaway: collect a sufficiently large rollout batch for stable gradients, but do not simply maximize it.
- Experiment/tuning factor: batch size should be tuned together with rollout length, number of agents, compute limits, and mini-batch count.

5. Other Important Implementation Details

- Parameter sharing: for homogeneous agents, the paper shares actor and critic parameters. This substantially improves learning in their SMAC ablations (e.g., very large gains on MMM2 and 6h_vs_8z).
- Heterogeneous agents: MPE Comm is the exception; because speaker/listener observation and action spaces differ, separate policies are used.
- Recurrent MAPPO: the released/appendix algorithm supports RNN/GRU actor and critic states, useful under partial observability because agents can retain information across timesteps.
- GAE + advantage normalization: used to generate a lower-variance policy-learning signal from rollout rewards.
- Value clipping, gradient clipping, orthogonal initialization, Adam optimization, and feature normalization are part of the implementation stack and should not be omitted casually when reproducing MAPPO.
- Death masking: in environments where agents can become permanently inactive/dead while the episode continues, the paper finds it beneficial to feed the critic a fixed zero-state plus agent ID for the dead agent. Relevant only if our environment includes such agent removal/failure states.

6. Limitations and Gaps

Gap / limitation	Why it matters
All tested tasks are cooperative	Does not establish behavior in mixed cooperative-competitive or general-sum settings.
Benchmark action spaces are discrete	Continuous-action multi-agent control is not established by these experiments.
Agents are homogeneous in almost all tests	Specialized/heterogeneous roles remain less studied; parameter sharing may not transfer directly.
Primarily empirical study	The paper gives strong practical evidence and hypotheses, but little formal explanation of why the recommended settings work.
No systematic emergent-communication study	The paper is a training/backbone reference, not a solution to learned messaging.
No channel realism	No latency, bandwidth limits, communication cost, message dropout/noise, or learned scheduling analysis.
Strong performance depends on implementation details	A weak 'PPO baseline' may be misleading if it omits the practices isolated by this paper.

7. What We Take From This Paper for Implementation

These are implementation lessons / things to consider from this paper, not a declaration of the project's final research direction.

- Use MAPPO as a serious cooperative-MARL baseline/backbone candidate, with IPPO retained as a meaningful decentralized baseline.
- Keep actors decentralized; if using MAPPO, allow the training critic access to a carefully designed global + agent-specific state.
- Normalize critic value targets using running statistics / PopArt-style normalization.
- Use conservative data reuse: on difficult tasks begin around 5-10 PPO epochs per fresh rollout rather than tens of epochs.
- Avoid excessive mini-batch splitting; begin with one mini-batch if memory allows, then tune.
- Keep PPO clipping conservative (paper suggests $\epsilon < 0.2$) and tune for stability vs convergence speed.
- Use sufficiently large rollout batches for reliable gradients without assuming 'larger is always better'.
- Use parameter sharing only where agents are genuinely homogeneous; do not force it across incompatible role/action spaces.
- Consider recurrent actor/critic networks if partial observability or delayed information requires memory.
- If agents can die/be disabled before an episode ends, explicitly test/implement death-state handling rather than letting critic inputs become inconsistent.

8. Experiments / Comparisons This Paper Makes Important

Comparison	Purpose
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IPPO vs MAPPO	Separates benefit of centralized critic information from basic PPO learning.
MAPPO critic: local vs EP/global vs AS/FP	Tests which critic information actually improves value learning.
Parameter sharing vs separate networks	Necessary when deciding whether homogeneous-agent sharing is beneficial.
5 vs 10 vs 15 training epochs	Checks sensitivity to rollout reuse / policy non-stationarity.
1 vs 2+ mini-batches	Checks whether additional gradient splitting destabilizes training.
Different clip ratios (<0.2 range)	Measures stability vs convergence speed.
Different rollout/batch sizes	Finds sufficient gradient quality without unnecessary sample/compute cost.
No-communication MAPPO baseline in any communication study	Prevents attributing gains from better centralized training to communication itself.
Multiple random seeds + held-out environment randomizations	Needed because MARL variance is high and memorized layouts would give misleading conclusions.

9. Paper-Level Bottom Line

The paper's central lesson is not that MAPPO is universally superior, but that a carefully implemented PPO baseline is much stronger and more sample-efficient in cooperative MARL than common assumptions suggested. The strongest practical evidence concerns training stability: value normalization, informative but compact centralized critic inputs, limited rollout reuse, conservative PPO clipping, appropriate batch size, and parameter sharing for homogeneous agents.

For an emergent-communication study, the paper therefore raises the baseline standard: communication should be evaluated against strong IPPO/MAPPO agents, and improvements must be separated from gains caused simply by centralized critic training or PPO engineering.

Reference

Yu, C., Velu, A., Vinitsky, E., Gao, J., Wang, Y., Bayen, A., & Wu, Y. *The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games*. NeurIPS 2022. arXiv:2103.01955v4 (04 Nov 2022).

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